Digital Privacy and Security

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Overview

- Threat modeling
- Encryption
- Digital architecture
- Legal jursidiction

Threat modeling

- Who might target you?
- What do they want?
- What are they capable of?

Transport encryption

- common, often enforced; HTTPS/TLS
- encrypts data in transit between you and the server
- server holds the keys
- useful for network interception, attackers posing as server
- not useful against requests from server; subpoena, CLOUD

At-rest encryption

- BitLocker, LUKS, FileVault
- common on modern smartphones, widely available on other devices
- data is encrypted when the device is powered off
- useful for lost/stolen devices, identity theft
- not useful for border/customs/police checkpoints, cloud-based data

End-to-end encryption

- encrypts communication between communicating users
- users hold the keys, server does not see the contents
- useful for keeping communications private
- not useful for preventing trusted users from sharing private messages
- the less data the company has, the less meainguflly they can comply with subpoena
- servers often still retain unencrypted access to metadata
  - phone numbers
  - usernames
  - timestamps

Zero-knowledge architecture

- E2EE + At-rest, and
- design philosophy that avoids giving server access to encryption keys
- only the user(s) ever have access to the keys
- Examples include Proton, Bitwarden, Signal
- easier to trust servers that cannot physically comply meaningfully with subpoena

Jursidiction

- Legal compulsion depends on who operates the service and where they are incorporated, not just where the data physically lives.
- US companies must produce data they control, regardless of physical server location (CLOUD Act)
- Warrantless collection of non-US persons' communications; includes gag orders (FISA, §702)
  - difficult to identify who is/is not a US resident; results in incidental collection of US-resident data with impuntiy
  - PRISM is the government's open back-door access to major tech companies data; different from legal subpoena mechanism

What to look for

- End-to-end encryption
- Zero-knowledge architecture
- Lack of “forgot your password” features are often a good sign
- Combination of jursidiction and technical defense

Self-audit

- What services do your teams rely on? List them out.
- What do those services have access to?
  - encryption keys, unencrypted data, metadata, etc.
- What jursidiction does the servicing organization fall under?

- If you are using high-risk services that could put you or others at risk, consider switching.

Slides and contact

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